

Linealidad de la integral

Teorema 1 (Linealidad de la integral). Sea (X, Σ, μ) un espacio de medida, $E \in \Sigma$ y $f, g: X \rightarrow \mathbb{R}$ funciones integrables en E (es decir, $f, g \in \mathcal{L}^1(\mu, E)$)

$$\implies \forall a, b \in \mathbb{R} : \int_E (af + bg) d\mu = a \int_E f d\mu + b \int_E g d\mu$$

Demostración: Veamos primero que $\forall a \in \mathbb{R} : \int af d\mu = a \int f d\mu$.

$$\text{por definición } \int af d\mu = \int (af)^+ d\mu - \int (af)^- d\mu$$

$$\text{Observamos que } \begin{cases} a \geq 0 \implies (af)^+ = af^+ \wedge (af)^- = af^- \\ a < 0 \implies (af)^+ = -af^- \wedge (af)^- = -af^+ \end{cases}$$

$$\implies \int (af)^+ d\mu - \int (af)^- d\mu = \begin{cases} a \int f^+ d\mu - a \int f^- d\mu & \text{si } a \geq 0 \\ -a \int f^- d\mu + a \int f^+ d\mu & \text{si } a < 0 \end{cases} = a \int f d\mu$$

$$\implies \int (af + bg) d\mu \stackrel{??}{=} \int af d\mu + \int bg d\mu = a \int f d\mu + b \int g d\mu$$

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